

Inventory Reorder Automation for Quick-Commerce

1. Problem Statement

Quick-commerce businesses manage a large number of products across multiple stores. Inventory requirements can vary depending on product demand, current stock levels, and supplier lead times.

The challenge is to identify products that may run out before replenishment arrives and determine which products need to be reordered first.

The goal of this project was to build an automated system that could analyze inventory data, identify stockout risks, calculate reorder requirements, prioritize products, and generate alerts for the operations team.

2. Objective

The project was built to:

- Monitor inventory across stores.
- Calculate how long current stock can support demand.
- Estimate demand during supplier lead time.
- Identify potential stockout risks.
- Calculate reorder quantities.
- Prioritize products based on inventory risk.
- Estimate reorder value and revenue at risk.
- Automate alerts and reporting.

3. Dataset

The project uses **500 SKU-store records** with 11 input columns.

Category	Fields
Product	Product ID, Product Name, Category
Location	Store, City

Category	Fields
Inventory	Current Stock
Demand	Average Daily Sales
Supplier	Lead Time, Supplier
Financial	Unit Cost, Selling Price

Each row represents one **product at one store**.

Data Quality

- Records: **500**
- Input columns: **11**
- Missing values: **0**

4. Approach

The analysis followed this workflow:

Inventory Data → Data Validation → Inventory Metrics → Risk Classification → Reorder Calculation → Recommendations → Alerts → Dashboard

Step 1: Inventory Coverage

I calculated **Days of Inventory** to understand how long the existing stock could support the current demand.

Days of Inventory = Current Stock ÷ Average Daily Sales

For example, Atta 5kg at Gurgaon-01 had:

- Current stock: 3
- Average daily sales: 10
- Inventory coverage: **0.30 days**

This indicated that the available inventory was very low compared with the current demand.

Step 2: Lead-Time Demand

I then considered supplier lead time because inventory risk depends not only on how much stock is available but also on how long replenishment takes.

Lead-Time Demand = Average Daily Sales × Lead Time

For example:

- Daily sales: 10
- Lead time: 2 days
- Lead-time demand: **20 units**

This helped compare expected demand during replenishment time with the available inventory.

Step 3: Reorder Calculation

Based on current inventory, demand, lead time, and target stock, I calculated the **reorder quantity** for products requiring replenishment.

The report also calculated:

- Reorder Value
- Current Inventory Value
- Revenue at Risk
- Reorder Required

This converted the analysis from simply identifying low stock into determining an actionable replenishment requirement.

5. Priority Classification

The inventory records were divided into four priority levels:

Priority	Count	%
URGENT	156	31.2%
HIGH	14	2.8%
MEDIUM	130	26.0%
LOW	200	40.0%
Total	500	100%

The **URGENT + HIGH** categories contained **170 records**, representing **34%** of the inventory analyzed.

This allowed the operations team to focus on the most important inventory issues first.

6. Example of an Urgent Inventory Issue

Rice 5kg — Gurgaon-02

Metric	Value
Current Stock	19
Average Daily Sales	8
Lead Time	4 days
Days of Inventory	2.38
Reorder Quantity	37
Reorder Value	₹12,950
Revenue at Risk	₹13,760
Priority	URGENT

The product had **2.38 days of inventory coverage**, while the supplier lead time was **4 days**.

This means the existing stock could potentially run out before the next delivery arrives.

The system therefore classified the product as **URGENT** and recommended a reorder of **37 units**.

7. Key Results

The analysis produced the following results:

KPI	Result
Inventory Records	500
Urgent	156
High Priority	14
Medium	130
Low	200
Stockout Risk	156
Reorder Required	353
Total Reorder Value	₹13.3L
Revenue at Risk	₹15.0L

KPI	Result
Average Days of Inventory	4.9 days

8. Key Insights

8.1 High replenishment requirement

353 out of 500 records, or **70.6%**, were identified as requiring a reorder.

8.2 Significant high-priority exposure

170 records were classified as either URGENT or HIGH priority.

8.3 Stockout exposure

156 records were identified as having stockout risk.

8.4 Financial exposure

The inventory analysis identified approximately **₱13.3L in reorder value** and **₱15.0L in revenue at risk**.

8.5 Lead time matters

The analysis showed that inventory coverage should be evaluated against supplier lead time rather than looking at current stock alone.

8.6 Store-level differences

Inventory requirements varied across stores, showing the importance of considering store-level demand and inventory conditions.

9. Automation

After completing the inventory analysis in Python, I automated the reporting and alert process.

Python

Used for:

- Data processing
- Inventory calculations
- Risk identification

- Reorder calculations
- Priority classification
- Recommendation generation

Google Sheets

Used for:

- Inventory Report
- Alerts
- KPI Dashboard

Google Apps Script

Used for:

- Reading inventory results
- Identifying priority items
- Creating alerts
- Sending email notifications

Automated Workflow

Python → Inventory Report → Google Sheets → Apps Script → Alerts → Email

10. Recommendation Output

The final Inventory Report includes an **AI Recommendation** field that converts the calculated inventory information into an easy-to-understand action.

For example:

URGENT: Reorder the recommended quantity immediately due to low inventory coverage and supplier lead time.

This makes the report more actionable for an operations user instead of requiring them to interpret every calculated metric themselves.

11. Dashboard

A KPI dashboard was created in Google Sheets to provide a quick overview of inventory health.

KPIs

- Total Inventory Records
- Urgent Items
- High Priority Items
- Reorder Required
- Stockout Risk
- Total Reorder Value
- Revenue at Risk
- Average Days of Inventory

Visuals

- Priority Distribution
- Reorder Value by Store
- Top Products by Revenue at Risk
- Inventory Risk Overview

The dashboard allows users to quickly identify **where inventory attention and financial exposure are concentrated**.

12. Outcome

The project transformed raw inventory data into an automated workflow for **inventory monitoring, prioritization, and replenishment planning**.

Instead of manually reviewing 500 records, the system highlights:

- Which products require attention
- Which products are at stockout risk
- How much inventory should be reordered
- How much the reorder may cost
- Which stores have greater exposure
- Which issues should be prioritized

The final solution combines **data analysis, business logic, reporting, and automation** in one workflow.